Chapter 13 | Written solutions

Structured State Spaces and Long Convolutions

Goal

Work through all eleven exercises, from a one-number memory to a changing state update. Track where each input is written, how often it is carried, and when it is read. The central checks compare a recurrence with its convolution, distinguish a compact state from the full history, and expose future leakage from an FFT that is too short. Exercises 9-11 are optional. Both PDF editions contain every explanation, result, and diagram; the notebook edition also includes the code and its recorded outputs.

1. Follow one state

Each step keeps one quarter of the old state, then adds the new input. Start with $x_0 = 0$. The state is read **after** adding the current input.

Position t	**Input** u_t	**Retained old state**	**Updated state** x_t
1	4	$(1/4)0 = 0$	4
2	0	$(1/4)4 = 1$	1
3	0	$(1/4)1 = 1/4$	$1/4$
4	2	$(1/4)(1/4) = 1/16$	$1/16 + 2 = 33/16$

Thus the states are $(4, 1, 0.25, 2.0625)$. The first input crosses three transitions before the fourth output, contributing $4(1/4)^3 = 1/16$. The newest input contributes 2 immediately, with lag zero.

For the shorter history $(4, 0)$, the states are $(4, 1)$. For $(0, 1)$, they are $(0, 1)$. Both leave **the same state, 1**, after two inputs. The state alone cannot distinguish those histories. Applying the same future inputs gives the same future states, even though the earlier outputs differed. A state can be sufficient to continue this recurrence without being a lossless record of the inputs.

2. Read by lag

An input enters with coefficient one. Each later transition multiplies its contribution by $1/4$, so the filter tap at lag ℓ is $k_\ell = (1/4)^\ell$. The first four taps are

$$(k_0, k_1, k_2, k_3) = (1, 1/4, 1/16, 1/64).$$

The first tap multiplies the current input, not the oldest input. For $u = (4, 0, 0, 2)$,

$$y_3 = k_0u_3 + k_1u_2 + k_2u_1 = 0 + 0 + (1/16)4 = 1/4,$$

$$y_4 = k_0u_4 + k_1u_3 + k_2u_2 + k_3u_1 = 2 + 0 + 0 + (1/64)4 = 33/16.$$

These match the third and fourth states. The recurrence applies a factor of $1/4$ at each intervening step; the filter already contains the product of those factors. A unit impulse, followed by zeros, therefore returns the filter itself.

3. Carry two features

Write the state as $(p_t, q_t)^\top$. The two coordinates receive the same input but retain different fractions of their previous values:

$$p_t = \tfrac{1}{2}p_{t-1} + u_t, \qquad q_t = \tfrac{3}{4}q_{t-1} + u_t, \qquad y_t = p_t - q_t.$$

Position	**Input**	**First coordinate p_t**	**Second coordinate q_t**	**Readout y_t**
1	4	4	4	0
2	0	2	3	-1
3	0	1	$9/4$	$-5/4$

The outputs are $(0, -1, -1.25)$. A zero input stops adding new information; it does not freeze the stored state. The first coordinate fades faster, so the readout becomes negative. The first output is zero because the two nonzero coordinates cancel. **A zero readout does not imply an empty state.** Later transitions can expose information that this readout currently cancels. A filter coefficient can likewise be zero or negative.

4. Choose a step size

The bilinear rule uses $A = -1$, $B = 1$, and the scalar identity 1. Substitute these before choosing Δ:

$$\overline{A} = \frac{1 - \Delta/2}{1 + \Delta/2}, \qquad \overline{B} = \frac{\Delta}{1 + \Delta/2}.$$

Step Δ	**Shared denominator**	**Retention $\overline{A}$**	**Write coefficient $\overline{B}$**
$1/2$	$5/4$	$(3/4)/(5/4) = 3/5$	$(1/2)/(5/4) = 2/5$
1	$3/2$	$(1/2)/(3/2) = 1/3$	$1/(3/2) = 2/3$

For zero new input, only the retention term remains. In one update, $\Delta = 1/2$ preserves more of the old state's magnitude because $3/5 > 1/3$. For instance, an old state of 15 becomes 9 or 5, respectively. These compare one step at each chosen step size, not equal durations of continuous time. The bilinear rule is a discretization; the recurrence and convolution agree on that discrete rule, not necessarily on the exact continuous-time solution.

5. Compare three calculations

The two-coordinate system in Exercise 3 writes the same input into both coordinates. The readout subtracts the second coordinate from the first. A first input of 4 gives equal coordinates, so the first readout is zero. Later, the two coordinates decay at different rates and no longer cancel. We will reproduce this behavior with a scan and with two ways of applying a filter.

Let the first input be u_1 and the initial state be $x_0 = 0$. The scan applies the update $x_t = \overline{A}x_{t-1} + \overline{B}u_t$ and then reads $y_t = Cx_t$. With inputs $(4, 0, 0)$, we get

t	**State** x_t	**Readout** y_t
1	$(4, 4)$	$4 - 4 = 0$
2	$(2, 3)$	$2 - 3 = -1$
3	$(1, 9/4)$	$1 - 9/4 = -5/4$

The filter takes a different view. Imagine a unit input followed by zeros. Its contribution to the state starts as $\overline{B}$ and passes through $\overline{A}$ once for every elapsed step. The filter coefficient at lag j is therefore $h_j = C\overline{A}^j\overline{B} = (1/2)^j - (3/4)^j$. The first three coefficients, in order of increasing lag, are 0, $-1/4$, and $-5/16$. Lag zero multiplies the current input. Each subsequent coefficient multiplies an input one step further in the past. The only nonzero input is 4 at $t = 1$, so the three outputs are $4h_0$, $4h_1$, and $4h_2$. This gives the same readouts as the scan.

Direct convolution computes those weighted sums in the time domain. The FFT method instead transforms the zero-padded input and filter, multiplies their spectra, and transforms the product back. The transform length is $3 + 3 - 1 = 5$, which accommodates the full linear convolution. The causal output keeps its first three entries. Both methods use the same filter coefficients.

The state trace has one row per input, and each row is the state **after** the corresponding update. The one-dimensional arrays for $\overline{B}$ and C both have length two. The helper multiplies $\overline{B}$ by one scalar input and takes the dot product of C with a two-coordinate state, yielding one scalar output per input. We compare float64 results with absolute tolerance 10^{-12} and relative tolerance zero. The recorded run agrees at every entry. An FFT implementation can introduce small roundoff differences, so the comparison uses the unrounded results and a tolerance.

6. Keep the tail in its place

A causal filter with three taps uses the current input and up to two earlier inputs. Number the input positions from zero in this exercise to track the indices in the FFT. With taps $h = (1, -1, 1)$ the linear convolution is $y_i = u_i - u_{i-1} + u_{i-2}$. Treat inputs outside positions 0 through 4 as zero. This gives the following full output.

Index i	**Calculation**	**Full linear output** y_i
0	$3 - 0 + 0$	3
1	$0 - 3 + 0$	-3
2	$1 - 0 + 3$	4
3	$0 - 1 + 0$	-1
4	$2 - 0 + 1$	3
5	$0 - 2 + 0$	-2
6	$0 - 0 + 2$	2

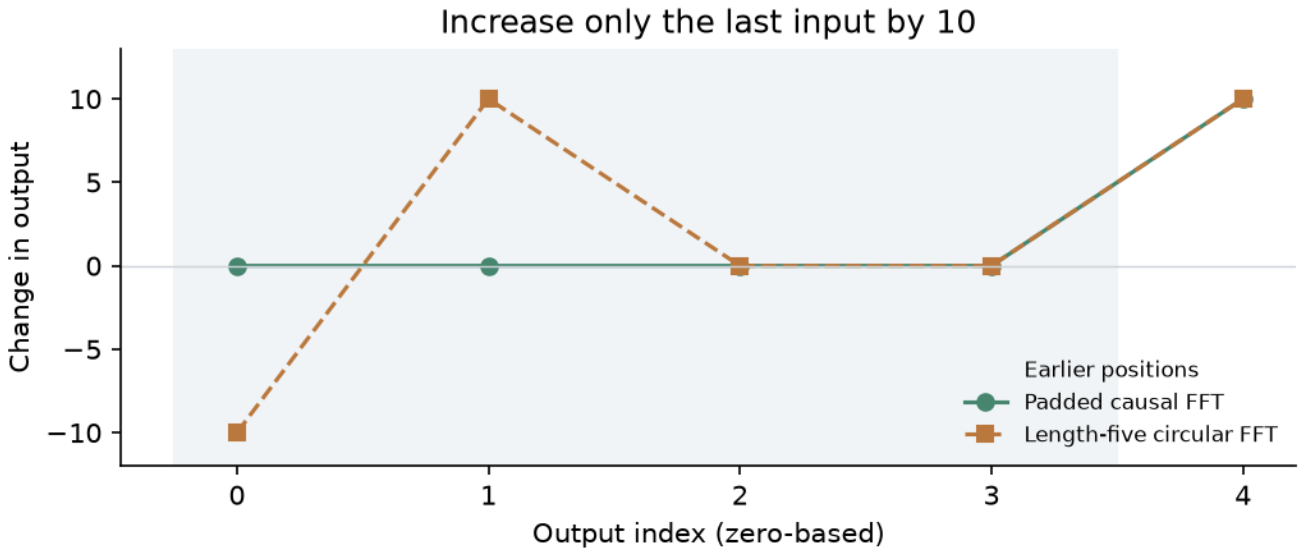
The last nonzero input at index 4 can still contribute two positions later through the tap at lag 2. In general the last possible output index is $(N-1)+(M-1)$ for input length N and filter length M. The minimum transform length that keeps all these indices separate is therefore $N + M - 1 = 5 + 3 - 1 = 7$. The provided FFT function pads to this length and returns the five outputs aligned with the input. We also run the length-seven transform directly to inspect the two tail entries it trims.

A length-five transform treats positions as periodic with period five. A product between an input at position j and a filter tap at lag r goes to the remainder of $j + r$ after division by five. For example, the term $2 \cdot (-1)$ from input 4 and lag 1 belongs at index 5 in the linear result. The short transform puts it at index 0. Likewise the term $2 \cdot 1$ from lag 2 folds from index 6 into index 1. The other three positions receive no folded term.

The circular result is thus $(3 - 2, -3 + 2, 4, -1, 3)$, or $(1, -1, 4, -1, 3)$. Passing `n=5` to each forward transform and the inverse gives all three calls the same period. The inverse also needs the explicit length because the half spectrum of three values could describe a real input of length four or five.

The last line isolates the extra contributions to the first two output positions. Both FFT checks use float64 inputs, zero relative tolerance, and an absolute tolerance of 10^{-12} to allow for roundoff. Rounding the printed rows only makes them easier to read. Cropping is safe after a padded linear convolution. Cropping cannot undo terms already folded into the output by the shorter transform.

Now increase the last input by 10. Correct causal outputs at indices 0 through 3 must stay unchanged: the changed input has not arrived at those positions. The causal difference is $(0, 0, 0, 0, 10)$. The too-short circular calculation instead changes by $(-10, 10, 0, 0, 10)$ because the new tail folds into indices 0 and 1. This tests causality directly, rather than relying only on agreement between two implementations.



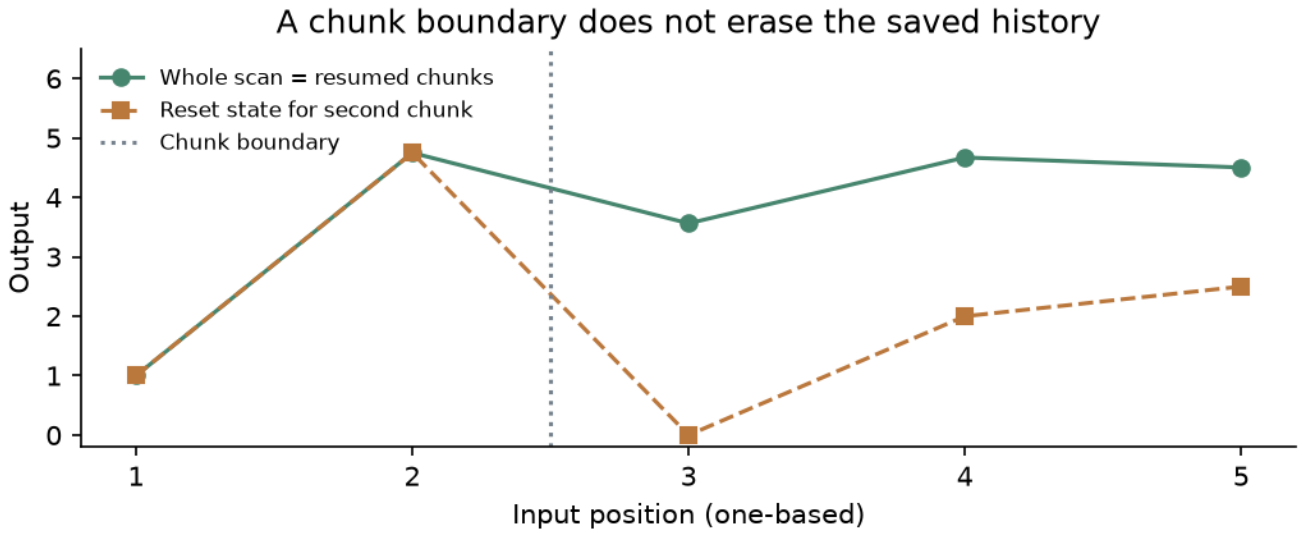
7. Resume from a saved state

Splitting the input into chunks does not change the recurrence. A chunk boundary is just a place to save the current state and resume later. For this one-coordinate system the update is $x_t = \frac{3}{4}x_{t-1} + u_t$, and $y_t = x_t$. The first two inputs give $x_1 = 1$ and $x_2 = \frac{3}{4} \cdot 1 + 4 = \frac{19}{4} = 4.75$. The last state of the first chunk, `first_states[-1]`, is a one-element vector containing x_2.

At the first input of the second chunk, the input is zero. Resuming gives $x_3 = \frac{3}{4} \cdot 4.75 = 3.5625$. Starting at zero gives $x_3 = 0$. The subsequent inputs 2 and 1 are identical in both runs, so their contributions cancel when comparing the states. The difference is multiplied by $3/4$ at each step.

Global step t	u_t	**Continue from** x_2	**Reset after** $t = 2$	**Difference**
3	0	3.5625	0	3.5625
4	2	4.671875	2	2.671875
5	1	4.50390625	2.5	2.00390625

In general, after j steps of the second chunk, the difference is $(3/4)^j x_2$ for this scalar system. The saved state preserves the effects of both earlier inputs in a single number. We can verify the joined output and the joined state trace independently.



8. Move a zero gate

Think of the input gate as controlling what gets written into the filter's history, and the output gate as controlling what gets read out at the current position. Number positions from one in this exercise, matching position 2 in the prompt. Let x_t be the input after its gate and z_t the filtered value. With taps $(1, 1/2, 1/4)$,

$$z_t = x_t + \frac{1}{2} x_{t-1} + \frac{1}{4} x_{t-2}.$$

Values of x_t before position 1 are zero. The output gate then multiplies z_t by its entry at position t. It acts after the filter, so its result does not get fed back into the filter.

If the input gate closes at position 2, it replaces the second input 2 with 0 before filtering. Then $x = (1, 0, 1, 1)$. If only the output gate closes there, the filter sees all of $x = (1, 2, 1, 1)$. Compute the two filtered sequences row by row.

Position t	**Filtered with input gate closed at 2**	**Filtered with all input gates open**
1	1	1
2	$0 + \frac{1}{2} = \frac{1}{2}$	$2 + \frac{1}{2} = \frac{5}{2}$
3	$1 + 0 + \frac{1}{4} = \frac{5}{4}$	$1 + 1 + \frac{1}{4} = \frac{9}{4}$
4	$1 + \frac{1}{2} + 0 = \frac{3}{2}$	$1 + \frac{1}{2} + \frac{1}{2} = 2$

With the zero on the input side, the output gate is all ones, so the final output equals the filtered sequence $(1, \frac{1}{2}, \frac{5}{4}, \frac{3}{2})$. With the zero on the output side, only the filtered value at position 2 is suppressed. The final output is $(1, 0, \frac{9}{4}, 2)$.

All these values are integers or binary fractions, so the float64 results can be checked for exact equality. To isolate the later effect of the closed input gate, compare the two **filtered** sequences, before either output gate acts.

At position 3, the original second input 2 would have contributed $2 \cdot \frac{1}{2} = 1$ through lag one. At position 4, it would have contributed $2 \cdot \frac{1}{4} = \frac{1}{2}$ through lag two. The closed input gate removes both contributions, so **positions 3 and 4** are affected. Position 1 is earlier than the changed input and cannot read it. The filter has no taps beyond lag two, so the effect would also be gone at position 5 if we extended the sequence.

Position 2 itself shows why the placement matters. Closing the input gate there leaves the earlier contribution $\frac{1}{2}$ from position 1. Closing the output gate there zeroes the whole filtered value, but the input at position 2 remains available to later filter calls.

9. Account for the initial state (optional)

With a nonzero initial state, the system can produce an output before any nonzero input arrives. Here the first input is zero, but the initial state $(1, -1)$ becomes $(1/2, -1/4)$ after the first update. Reading with $C = (2, 1)$ gives $2(1/2) - 1/4 = 3/4$. A zero-state convolution gives zero at that step and needs a separate term for the initial state.

For each input, halve the first coordinate of the previous state, quarter the second, and add the same input to both coordinates. Then read twice the first coordinate plus the second.

t	**Input** u_t	**State** x_t	**Output** Cx_t
1	0	$(0.5, -0.25)$	0.75
2	3	$(3.25, 2.9375)$	9.4375
3	1	$(2.625, 1.734375)$	6.984375
4	0	$(1.3125, 0.43359375)$	3.05859375

Linearity lets us separate the response into two parts. The zero-state part follows the inputs starting from 0. The initial-state part follows x_0 with all inputs set to zero. At step t the latter has applied the transition t times. Applying the readout to both parts gives

$$y_t = \sum_{j=0}^{t-1} h_j u_{t-j} + C\overline{A}^t x_0, \qquad h_j = C\overline{A}^j \overline{B}.$$

The filter starts at lag zero because the current input is read immediately. The correction starts at $t = 1$ because the first returned output is read after the first transition. Here $h_j = 2(1/2)^j + (1/4)^j$, giving taps $(3, 1.25, 0.5625, 0.265625)$. For the correction, the two coordinates of $\overline{A}^t x_0$ are $(1/2)^t$ and $-(1/4)^t$. Reading them gives $2(1/2)^t - (1/4)^t$.

t	**Zero-state convolution**	**Initial-state correction**	**Sum**
1	0	0.75	0.75
2	9	0.4375	9.4375
3	6.75	0.234375	6.984375
4	2.9375	0.12109375	3.05859375

The two outputs agree within the float64 absolute tolerance 10^{-12} with relative tolerance zero. If we started the powers at zero, the first correction would be $Cx_0 = 1$, which reads the state before processing u_1. The scan returns the output after the update, so the correct first correction is $3/4$.

10. Count streaming storage (optional)

Count numerical history carried across a boundary before the next input arrives, per feature, once the direct filter's history is filled.

Filter length in direct method	**SSM with 16 state coordinates**	**Direct convolution history**
1024	16 numbers	$1024 - 1 = 1023$ numbers
2048	16 numbers	$2048 - 1 = 2047$ numbers

The next incoming sample supplies lag zero. The direct implementation therefore retains the $L - 1$ preceding samples, not L past samples. Doubling L adds 1024 carried values; the SSM's chosen state width stays 16. Filter taps and transition weights are parameters, while temporary products and sums are workspaces. They are excluded by the question. The teaching helper's full state trace is also excluded: saving that trace for n inputs would require $16n$ values.

Smaller storage does not establish identical outputs for an arbitrary filter. The SSM can realize filters of the form $k_\ell = C\overline{A}^\ell \overline{B}$ with its chosen 16-coordinate state. That structured family need not contain a given arbitrary 1024-tap filter. Exact recurrence–convolution equivalence applies to a filter generated by the SSM itself. This storage comparison does not claim equal expressive power or predict a runtime speedup.

11. Check a changing rule (optional)

Since $x_0 = 0$, the first state is $x_1 = u_1$, regardless of a_1. Substituting the earlier states gives

$$y_2 = \tfrac{1}{2}u_1 + u_2, \qquad y_3 = \tfrac{3}{4}(\tfrac{1}{2}u_1 + u_2) + u_3 = \tfrac{3}{8}u_1 + \tfrac{3}{4}u_2 + u_3.$$

The coefficient of u_1 in y_2 is $1/2$. The coefficient of u_2 in y_3 is $3/4$. Both have lag one, but cross different transitions. One fixed lag-one tap cannot equal both values. The coefficient of u_1 in y_3 is the product $a_3 a_2 = 3/8$.

For these prescribed factors, the whole map is multiplication by

$$\begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/8 & 3/4 & 1 \end{pmatrix}.$$

It is still **linear in the inputs**, because its coefficients do not depend on them. It is not time invariant: the first subdiagonal is not constant. A fixed convolution would have the same tap along each subdiagonal. If the retention factors are instead computed from the inputs, the resulting complete map is generally nonlinear as well. These are two different properties. Losing time invariance alone is enough to lose the one-filter representation.

Checks

All eleven exercises have complete solutions. The executed checks cover state collisions, lag indexing, cancellation in the readout, discretization, scan/direct/FFT agreement, circular wraparound, future perturbations, chunk continuation, gate placement, initial-state corrections, storage counts, and linear but time-varying updates. Floating-point comparisons use explicit tolerances; plots display the computed arrays.

Next Steps

Change one retention factor or one gate and predict which later outputs will change before rerunning. Keep a filter's lag indices separate from the sequence's absolute positions. The next chapter lets the input control the state update itself, so a single fixed lag filter will no longer describe the complete sequence.